

Task 2: Language Modelling

Data Processing & Methodology:

I trained a word-level language model using Agatha Christie's *"Poirot Investigates"*. The text was cleaned (converted to lowercase, extra whitespace removed) and tokenized into a vocabulary of unique words. I created a dataset using a sliding window approach with a sequence length of **40 words**.

Instead of a traditional RNN/LSTM, I implemented a **Transformer-based architecture**. The model features an Embedding layer (dim=256), **Positional Encoding** to retain sequence order, and a Transformer Encoder with **4 layers** and **8 attention heads**. This architecture was chosen for its superior ability to handle long-range dependencies compared to LSTMs. The model was trained using **CrossEntropyLoss** and the **AdamW** optimizer (lr=3e-4) for 15 epochs.

Results & Text Generation:

The training loss converged to **0.16**, indicating the model successfully learned the statistical patterns of the source text. When prompted with specific seeds, the model generates text stylistically similar to the author. For example:

Prompt: "Mon ami,"

Output: "Mon ami, said poirot, this blasted old man... he'll get me if i stop here. The young man wiped the perspiration from his brow."

Demo:



The screenshot shows a Jupyter Notebook with a dark theme. At the top, there's a tab labeled 'Demo Mode'. Below it, a code cell is expanded, showing the following Python code:

```
[1] # Run Task 2 Demo
    input_prompt = input("Enter prompt: ")
    print(generate_demo(input_prompt))
```

Below the code cell, the output is displayed, showing a prompt and its corresponding generated text:

```
... Enter prompt: "Mon ami," said Poirot,
    said poirot, with a gallant little bow, as the children were removed reluctantly by a stern nurse. lady yardly smoothed her ruffled hair. "i adore them
```

Comparison Between Custom Transformer LLM and ChatGPT

Aspect	Custom Transformer Model	ChatGPT (LLM)
Training Data	Trained only on <i>Poirot Investigates</i>	Trained on extremely large, multi-domain corpus
Model Size	4-layer Transformer (256-dim embeddings)	Billions of parameters, deep architecture
Context Handling	Limited to 40-token sequences	Handles long-range context (thousands of tokens)
Text Coherence	Locally coherent, occasional grammatical drift	Strong global coherence and logical flow
Creativity & Generalisation	Recombines learned phrases from dataset	Generates novel, stylistically consistent content

The custom Transformer language model successfully generates text in a style like Agatha Christie, demonstrating that it has learned local word dependencies and stylistic patterns from the training corpus. However, because it was trained on a single book with a limited context window and relatively small parameter size, it occasionally loses grammatical structure and long-range semantic consistency. In contrast, ChatGPT benefits from significantly larger model capacity and exposure to vast and diverse datasets, enabling it to maintain coherence over longer passages and generate more original yet stylistically appropriate text. The differences primarily arise from model scale, training data diversity, and contextual modelling capacity.